

DOCUMENT: SOLUTION REQUIREMENTS (FR & NFR)

Phase	Project Design Phase II
Document	Solution Requirements (Functional & Non-Functional)
Project Name	PetPaws – Pet Foods & Accessories Store
Team ID	[TEAM-ID]
Date	14 June 2026
Maximum Marks	4 Marks

Functional Requirements

The following functional requirements define all the features and capabilities that the PetPaws application must provide to its users and administrators.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration & Authentication	Registration via email and password form JWT token generation on successful registration Registration via Gmail (future) Password hashing with bcryptjs (salt rounds: 10)
FR-2	User Login & Session Management	Login via email and password JWT token stored in localStorage Axios interceptor auto-attaches Bearer token Auto-logout on 401 Unauthorized response Admin login via separate endpoint (POST /api/auth/admin/login)
FR-3	Product Catalog Management	Browse all products with pagination (12 per page) Filter by category (15 types), pet type (8 types), price range, brand, minimum rating Sort by newest, most popular, top rated, price asc/desc Full-text keyword search using MongoDB text index View product detail with image gallery, variants, reviews View related products from same category
FR-4	Shopping Cart	Add product to MongoDB-persisted cart Update item quantity (capped at stock) Remove individual items from cart Clear entire cart Real-time cart badge count on Navbar Automatic inactive product cleanup on cart fetch Free shipping calculation (free above ₹500) 5% GST tax computation
FR-5	Wishlist Management	Add and remove products from wishlist View wishlist page with all saved products Move product from wishlist to cart in one click Persistent wishlist stored on User document in MongoDB

FR-6	Multi-Step Checkout	Step 1: Enter or select saved shipping address Step 2: Choose payment method (COD / Card / UPI / Net Banking) Step 3: Review full order before placement Server-side stock validation for every item before order creation Automatic stock deduction and soldCount increment on successful order Cart cleared after successful order placement
FR-7	Order Management (Customer)	View order history with status filter tabs View order detail with status timeline (Pending → Confirmed → Processing → Shipped → Delivered) Cancel order (Pending or Confirmed status only) Stock restoration on order cancellation View tracking number when assigned by admin Order confirmation page with Order ID after placement
FR-8	Product Reviews & Ratings	Submit star rating (1-5), review title and comment One review per user per product (compound unique index) Auto-assign Verified Purchase badge when user has a Delivered order for the product Recalculate product ratings.average and ratings.count after each review add/delete Display reviews sorted by newest first with reviewer avatar
FR-9	User Profile Management	Update display name and phone number Change password with current password verification Add, edit, delete saved delivery addresses Set default address (auto-applied at checkout) View wishlist from profile context
FR-10	Admin Dashboard & Analytics	KPI cards: Total Revenue, Total Orders, Total Customers, Total Products Monthly revenue bar chart using MongoDB aggregation pipeline Low-stock alert table (< 10 units) Recent orders quick-view table (last 5) Top-selling products ranking by soldCount
FR-11	Admin Product Management	Create product with full attribute set (name, description, brand, weight, price, discount, stock, category, petType, images, tags, featured, active) Edit product with live discounted price preview Soft-delete product (isActive: false) Search and filter product table Toggle Featured Product status for homepage display
FR-12	Admin Order Management	View all orders with status filter tabs (20 per page) Update order status through full lifecycle Add courier tracking number to order Add internal status notes recorded in statusHistory array View customer details and full order breakdown
FR-13	Admin Customer Management	View paginated customer list (20 per page) Search customers by name or email Activate or deactivate customer accounts View customer join date, last login, address count
FR-14	Admin Store Settings	Configure store name, tagline, contact email and phone Set shipping fee and free shipping threshold Set GST/tax

		rate Add and delete homepage banner images with title, subtitle, and link View configured product categories with active status
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Non-Functional Requirements

The following non-functional requirements define the quality attributes and constraints that govern the design and operation of the PetPaws platform.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	Responsive mobile-first CSS design ensures usability across desktop, tablet, and mobile viewports. Intuitive 3-step checkout wizard with progress indicators. Non-blocking toast notifications (react-hot-toast) provide real-time feedback. Admin sidebar collapses to icon-only mode for maximising content workspace. Empty states (empty cart, empty wishlist, no orders) display actionable guidance rather than blank pages.
NFR-2	Security	All passwords are bcrypt-hashed with salt rounds of 10 before persistence. JWT tokens are signed with a configurable secret (JWT_SECRET) and expire in 7 days (JWT_EXPIRE). Password field has select: false in Mongoose schema — never returned in API responses. The protect() middleware verifies JWT on every private route. The adminOnly() middleware enforces role-based access control. Deactivated accounts receive 401 on any login attempt. CORS is configured to permit requests only from the React client origin.
NFR-3	Reliability	All API controllers use try-catch blocks with structured JSON error responses. Stock validation is performed server-side before order creation to prevent overselling. Soft-delete (isActive: false) for products preserves order history integrity. StatusHistory array provides an immutable audit trail for every order state transition. Compound unique index on (user, product) in Review schema prevents duplicate review submissions.
NFR-4	Performance	Product listing is paginated at 12 products per page to reduce response payload size. MongoDB text index on (name, description, brand, tags) enables sub-50ms full-text keyword searches. Admin dashboard uses MongoDB aggregation pipeline to compute KPIs and revenue charts server-side, avoiding large data transfers to the client. Vite build tool provides fast HMR (< 100ms) in development and optimised production bundles with code splitting.
NFR-5	Availability	The three-tier MERN architecture decouples the frontend (Vercel), backend (Railway/Render), and database

		(MongoDB Atlas) — each layer can restart independently without affecting others. MongoDB Atlas provides 99.995% uptime SLA with automatic failover across replica set members. The stateless JWT authentication model allows the API to be deployed across multiple instances behind a load balancer without shared session state.
NFR-6	Scalability	MongoDB's document model and Atlas auto-sharding support horizontal scaling for growing product and order volumes without schema migrations. The RESTful API architecture allows individual microservice decomposition (auth, products, orders, admin) as traffic grows. React Context API state management can be upgraded to Redux Toolkit for enterprise-scale client-side state without refactoring the component tree. Environment-driven configuration (.env) enables seamless promotion from development to staging to production.